

BRIDGEPULSE · DESIGN DECISION RECORD

BridgePulse recommendations accepted

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DOCUMENT CONTROL

Document ID	BRP-DDR-002	Version	0.1
Status	Draft	Date	2026-09-25
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/bridgepulse		
Source	docs/decisions/0002-recommendations-accepted.md @ 520f6f8		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002); record every newly decided item, what changed in the repo and the items still open

0002: Recommendations accepted

- **Date:** 2026-09-25
- **Status:** accepted. Every item below that carried a recommendation is decided by Amish, 2026-09-25: go with recommendation. Items without a recommendation remain "Proposed, awaiting Amish".

■ Context

On 2026-09-25 Amish wrote: "i accept all your recommendations, go with them across all repos." Before that, the TRL 2 review items were recorded in BRP-DDR-001 as adopted for TRL 3 work, open for his review, and the TRL 3 review (docs/REVIEW.md , session 2026-09-25, TRL 3) added four new items. Where a recommendation offered several options, the recommended option is the decision. The portfolio stays at TRL 3; TRL 4 is on hold by Amish's instruction, so any decided item that needs a build, test, measurement or purchase is recorded as decided but on hold.

■ Options considered

The options for each item are those in docs/REVIEW.md (sessions 2026-09-25, /populate and TRL 3) and in BRP-DDR-001.

■ Decision

Table 1. Items decided by Amish, 2026-09-25: go with recommendation.

#	Item	Decision	What changed in the repo
D1 to D11	BRP-DDR-001 items (budget and R12, ambient excitation, hourly windows, hub on girder and FieldNode on post, foil half-bridges, one accelerometer, RP2040 class controller, engineer-only alerts, temperature-compensated trend, shared lab components, no change to pitch or problem)	As recommended in BRP-DDR-001	Status wording in BRP-DDR-001 (v0.2), BRP-PRB-001, BRP-PRC-001, BRP-REQ-001 and <code>bom/bom-notes.md</code> . <code>budget_usd</code> stays at \$250, which covers the BridgePulse-specific parts (D1: the recommendation redefined what the budget covers, applied to R12 at v0.3; no new figure). Pitch and problem unchanged (D11)
D12	R2 and people's mass (TRL 3 review item 4)	Options (a) and (b) together: gate out record segments while someone is on the span, found from the strain step or acceleration envelope, and regress each hourly frequency on a load indicator in TwinKit; R2 stays at 0.2 % until the method is checked on recorded data	Firmware rule in BRP-PRC-001 v0.4 (gate from the strain step with 0.25 s either side; accept a fit only if the peak is 10 dB above noise with plausible damping; otherwise report "no clear peak"). BRP-CAL-001 v0.2 simulates it: scatter 0.015 % or less against 1.5 to 3.1 % ungated. R2 moves from not met to at risk; target unchanged (BRP-REQ-001 v0.4). The check on recorded footfall is TRL 4: decided, on hold. The regression runs in TwinKit (cross-repo action)
D13	Reduced payload (TRL 3 review item 5)	Adopt an 11-byte summary for data rates limited to 11 bytes (US915 DR0, AS923 DR2 with dwell time), used once the region is known	R7 restated in BRP-REQ-001 v0.4; summary defined in BRP-PRC-001 v0.4 and BRP-CAL-001 v0.2, section G (first frequency, its amplitude, two strain means, steel temperature, status with gated seconds; 371 ms at SF10). The hub picks it when FieldNode reports an 11-byte limit, so R7 is met on paper whichever region is chosen. The region itself stays open in FieldNode
D14	Cold-range temperature (TRL 3 review item 6)	Ice-point check plus a probe specified over the full range, since one fits the cost margin	Item 6 is now a TMP1826 class 1-Wire probe potted in a stainless sheath, ± 0.3 °C from -40 to $+105$ °C on the maker's page, \$6.00 each (was a DS18B20 class probe at \$4.00). R5 verification adds an ice-point check; R5 moves from at risk to met on paper. BOM, BOM notes, BRP-PRC-001 v0.4, BRP-CAL-001 v0.2 (section E) and drawing BRP-DWG-001 Rev P2 (probe note) updated. The ice-point and CalRig checks are TRL 4: decided, on hold
D15	Cost margin (TRL 3 review item 7)	No change now; if quotes come in higher, the first savings are a consumer high-endurance card and a cheaper accelerometer breakout that still meets R1	Recorded in <code>bom/bom-notes.md</code> and BRP-CAL-001 v0.2, section K. With D14 the BridgePulse-specific parts are \$248.00 (was \$244.00), a \$2.00 margin (was \$6.00); complete monitor \$374.00 (was \$370.00). Obtaining quotes is purchasing (TRL 4): on hold

Table 2. Items that remain open.

#	Item	Status
O1	FieldNode sensor port pinout and protocol. No recommendation was made; the FieldNode candidate pinout remains a working assumption	Proposed, awaiting Amish and the FieldNode project
O2	First host bridge and co-design partner. No preference stated	Proposed, awaiting Amish
O3	Data ownership and what is published beyond monitoring status. No recommendation was made	Proposed, awaiting Amish
O4	LoRaWAN region for the first installation (FND-DDR-001, O1). No recommendation in this repo; D13 makes R7 independent of it	Proposed, awaiting Amish

■ Consequences

- Requirement status (BRP-CAL-001 v0.2): 0 not met, 3 at risk (R2, R10, R12), 3 not verifiable at TRL 3 (R3, R4, R8), 4 met on paper (R5, R6, R7, R9), 3 met by design (R1, R11, R13). Before: 1 not met, 4 at risk, 3 not verifiable, 2 met on paper, 3 met by design.
- `project.yaml` : only the evidence list gains this record; `budget_usd` stays at \$250, with `trl: 3` and `trl_target: 3`.
- Cross-repo actions, listed in `docs/REVIEW.md` : TwinKit to host the load regression of D12; FieldNode to review mounting its V-block clamps on a 50 mm square handrail post and to report the payload limit of the data rate in use to the sensor port (D13); FieldNode pinout (O1). No other repo is edited here.
- TRL 4 remains on hold by Amish's instruction. Nothing in this record authorizes building, testing, measuring or purchasing.